潘径浩 2023080041

S 11-10 "膜片涂黑"

1. 基尔霍夫定律 ⇒ 给定温度下，$\varepsilon = \alpha$

（发射率 = 同温度黑体辐射吸收率）

换言之，吸收能力强（$\alpha\uparrow$），向外辐射能量能力强（$\varepsilon\uparrow$）

2. $E = \varepsilon\sigma T^4$，涂黑后的表面发射率（$\varepsilon$）> 原色，相同T，辐射率E↑

~~S 11-13~~

Q 11-3 $t = 2000℃$, $T = 2273K$,

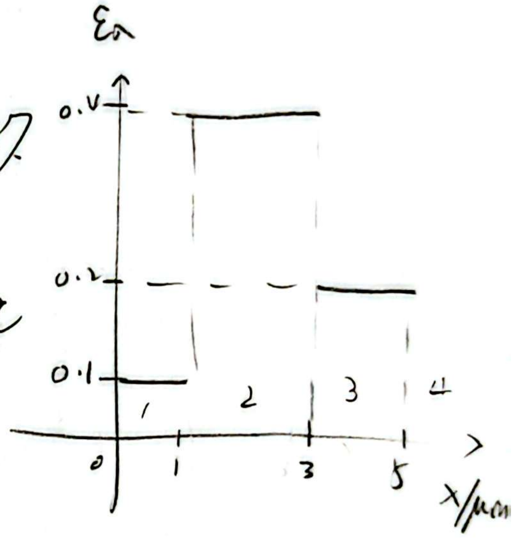
则处中呈仔, 以 $\lambda_{可见光} \times 10^{6} \in (0.38, 0.76)$

$\lambda_1 T = 0.38 \times 2273 = 863.74\ \mu m K$

$\lambda_2 T = 0.76 \times 2273 = 3172.5\ \mu m K$

查表, $\begin{cases} F_b(0-\lambda_1 T) = 0\% \\ F_b(0-\lambda_2 T) = 3.16\% \end{cases}$

∴ (份额) $F_b(\lambda_1 T-\lambda_2 T) = 3.16\%$



Q 11-6 漫射物体, $t_{surface} = 1205℃$

求 ε, E
(发射率) (辐射力)

∴ $T_{surface} = 1473\ K$

$\begin{cases} \lambda_1 T = 1\mu m \times T_{surface} = 1473\ \mu m K \\ \lambda_2 T = 3\mu m \times T_2 = 4419 \\ \lambda_3 T = 5\mu m \times T = 7365 \end{cases}$

查表, F_b

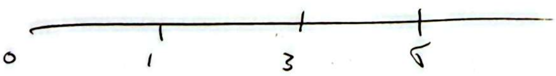
区间 1, 2, 3, 4

$0 \sim 1\mu m$: 1.15%

$0 \sim 3\mu m$: 55.22%

$0 \sim 5\mu m$: 82.75%

$\begin{cases} \varepsilon_1 = 0.1 \\ \varepsilon_2 = 0.4 \\ \varepsilon_3 = 0.2 \\ \varepsilon_4 = 0 \end{cases}$



∴ $\varepsilon = (0.1)(1.15-0) + (0.4)(55.22-1.15) + (0.2)(82.75-55.22)$

$= 27.249\%$

Q11-8 $\quad T = 1400K$,

$\because E_b = \sigma_b T^4$, $\sigma_b = 5.67 \times 10^{-8}$

$= (5.67 \times 10^{-8}) \times 1400^4 = 217819 \ W/m^2$

$\because \quad \lambda_{max} = \frac{\lambda_{max} T}{T} = \frac{2897.6}{1400} \approx 2.07 \mu m$